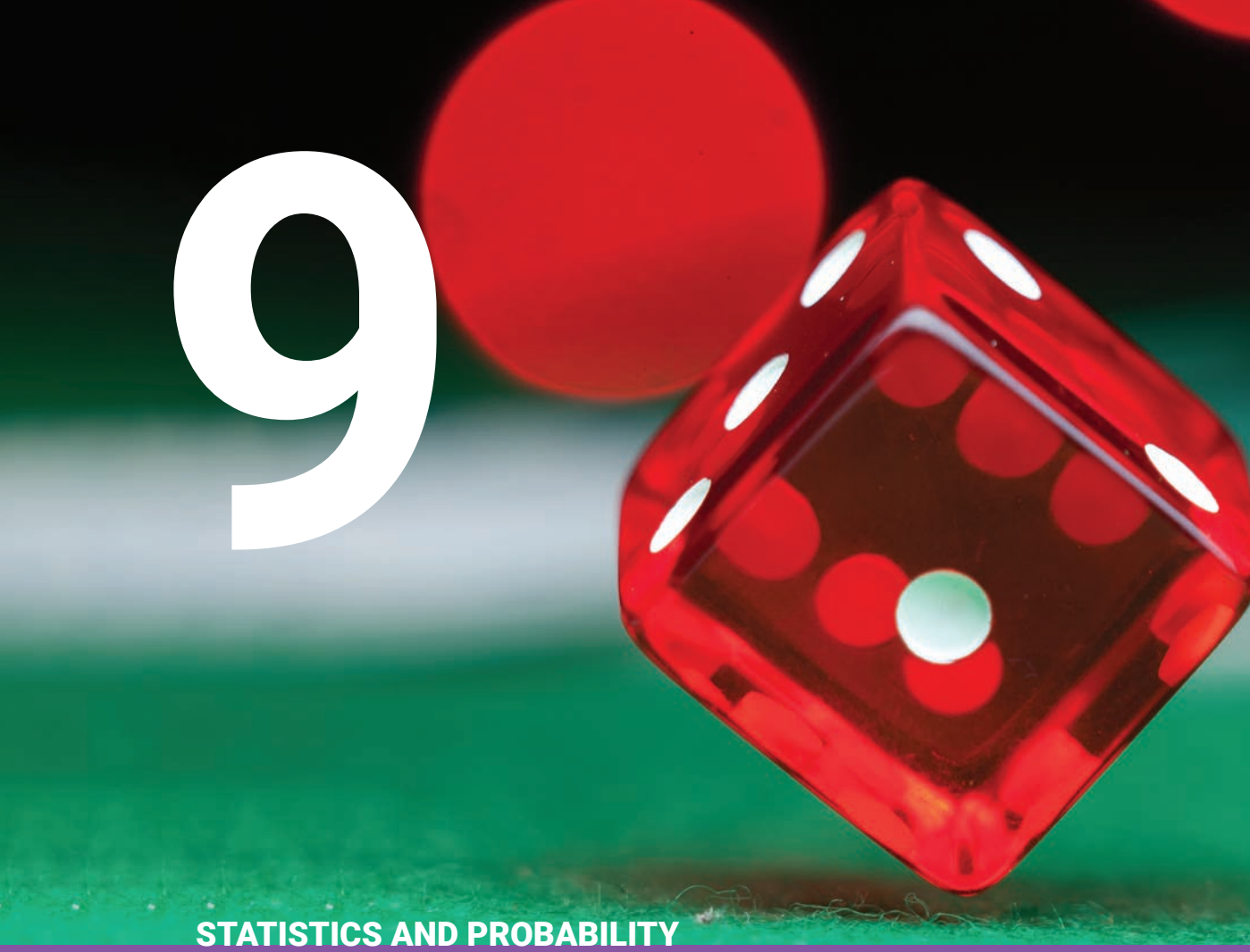


# 9

STATISTICS AND PROBABILITY

# PROBABILITY

Probability theory, the study of chance, began in the 17th century when 2 great mathematicians, Blaise Pascal and Pierre de Fermat, wrote to each other to discuss mathematical problems arising from games of chance. Since then, probability has become an essential branch of mathematics that is used widely in fields such as weather forecasting, finance, insurance, politics and medical testing.




## Chapter outline

| | | Working mathematically | | | | |
|---|---|---|---|---|---|---|
| **9.01** | Observed probability | U | F | PS | R | C |
| **9.02** | Sets and Venn diagrams* | U | F | | R | C |
| **9.03** | Two-way tables* | U | F | | R | C |
| **9.04** | Tree diagrams | U | F | PS | R | C |
| **9.05** | Selecting with and without replacement | U | F | PS | R | C |
| **9.06** | Dependent and independent events | U | F | PS | R | C |

*PATH

## Wordbank

**relative frequency** The frequency of an event over repeated trials as a fraction of the total number of trials

**set** A collection of distinct objects, for example, set $X = \{1, 2, 3, 4, 5, 6\}$

**simulation** An artificial imitation of a chance experiment, usually using technology, to approximate probabilities of events

**tree diagram** A diagram of branches for listing all the possible outcomes of a multi-step chance experiment

**trial** One go or run of a repeated probability experiment, for example, one roll of a die

**two-step experiment** A chance experiment with 2 steps or stages, such as rolling a pair of dice

**two-way table** A way of grouping items into 2 overlapping categories, such as gender and the ability to drive a car

**Venn diagram** A diagram of circles (usually overlapping) for grouping items into categories

**Quiz**
Wordbank 9

**U** = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

### Videos (10):

**9.01** Experimental probability
**9.02** Venn diagrams 1 • Venn diagrams 2
**9.03** Two-way tables 1 • Two-way tables 2
**9.04** Tables and tree diagrams • Tree diagrams 3 • Tree diagrams 1 • Tree diagrams 2
**9.05** Selecting with and without replacement

### *Twig* videos (6):

**9.01** Freak waves
**9.06** The Monty Hall problem • The birthday paradox • The card counter • The prisoner's dilemma • Bayesian robots

### *PhET* interactive (1):

**9.01** Plinko probability

### Quizzes (5):

- Wordbank 9
- SkillCheck 9
- Mental skills 9
- Language of maths 9
- Test yourself 9

### Worksheets (9):

**SkillCheck** Probability review
**9.01** Relative frequencies • Coins probability • Dice probability
**9.03** Two-way tables • Two-way probability tables
**9.04** Tree diagrams
**9.05** Multi-step experiments
**Mind map** Probability

### Puzzles (7):

**9.01** Dice probability
**9.02** Venn diagrams matching activity • And/or problems • Venn diagrams
**9.03** Combined events: Two-way tables
**9.04** Combined events: Tree diagrams
**Language of maths** Probability crossword

### Technology (1):

**9.01** Long-run proportion

### Spreadsheet (1):

**9.01** Long-run proportion

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

## In this chapter you will:

- calculate expected and observed probabilities, including from simulated chance experiments
- (PATH) describe compound events using terminology such as 'and', exclusive 'or', inclusive 'or' and 'at least'
- (PATH) use sets, set notation, Venn diagrams and two-way tables to represent sample spaces and compound events to solve probability problems
- use tree diagrams and tables (arrays) to represent the sample space of two- and three-step chance experiments, with and without replacement, to solve probability problems
- identify dependent and independent events, and use the product rule for independent events

## SkillCheck ANSWERS ON P. 615

**1** If a die is rolled, which of the following is more likely? Select the correct answer **A**, **B**, **C** or **D**.

**A** a number less than 3 **B** an even number

**C** a number that is 3 or more **D** a number that is a prime number

**2** A bag contains 5 10-cent coins, 4 20-cent coins and 3 50-cent coins. A coin is drawn at random from the bag.

**a** How many outcomes are in the sample space?

**b** Are the outcomes equally likely?

**3** For this spinner, the red sector is twice as large as each of the other sectors. Find the probability that when the spinner is spun, the arrow lands on:



**a** red

**b** purple or blue

**c** not green.

**4** A normal die is rolled. What is the probability of rolling:

**a** a 7? **b** a number less than 7?

**5** The probability of Danica hitting a bullseye when playing darts is 0.6. What is the probability of Danica not hitting the bullseye?

**6** Aditi bought 10 tickets in a raffle in which 400 tickets were sold and there is only one prize. What is the probability that Aditi will win the prize? Select **A**, **B**, **C** or **D**.

**A** $\frac{1}{400}$ **B** 2.5% **C** 0.0025 **D** 0.975

**7** The probability of cloudy skies tomorrow is 0.85. What is the probability of clear skies?

Quiz
SkillCheck 9

Worksheet
Probability review

9.01

# Observed probability 9.01

Probability calculated using the formula:

$$P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

is more specifically called **theoretical probability** or **expected probability**.

We can also determine probability based on the results of a trial that has been repeated many times, such as testing the effectiveness of 100 light globes, or rely on past statistics, such as weather patterns or the ages of drivers having car accidents. This type of probability is called **observed probability** or **experimental probability**, which is based on **relative frequency**, the number of times an **event** occurred as a fraction of the total frequency of outcomes.

Videos
Experimental probability
Freak waves

**Worksheet**
Relative frequencies

**Puzzle**
Dice probability

**Interactive**
Plinko probability

**Technology**
Long-run proportion

**Spreadsheet**
Long-run proportion

## Observed probability

$$P(E) = \frac{\text{number of times the event happened}}{\text{total number of trials}}$$

$$\text{or } P(E) = \frac{\text{frequency of } E}{\text{total frequency}}$$

**Expected frequency** is the expected number of times an event will occur over repeated trials.

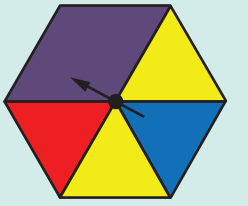
## Expected frequency

Expected frequency = theoretical probability × number of trials.

## Example 1

This spinner was spun 160 times and the results are shown.

| Outcome | Frequency |
|---|---|
| Red | 30 |
| Blue | 32 |
| Yellow | 50 |
| Purple | 48 |



**a** Calculate, as a decimal:

**i** the observed probability that the arrow stops on blue

**ii** the theoretical probability that the arrow stops on blue.

**b** Are the observed and theoretical probabilities similar?

**c** If the spinner is spun 500 times, calculate the expected frequency of spinning purple based on the theoretical probability.

### Solution

**a i** Observed probability:

$$P(\text{Blue}) = \frac{32}{160}$$
$$= 0.2$$

**ii** Theoretical probability:

$$P(\text{Blue}) = \frac{1}{6}$$
$$= 0.1666...$$
$$\approx 0.17$$

**b** By comparing the decimals for the 2 answers, we see that the observed and theoretical probabilities are similar.

**c** Expected frequency of purple $= \frac{1}{3} \times 500$ probability × number of trials

$$= 166.6666...$$
$$\approx 167$$


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## Example 2

James rolled a die 100 times and recorded the results in a table.

| Outcome | Frequency |
|---|---|
| 1 | 23 |
| 2 | 19 |
| 3 | 11 |
| 4 | 12 |
| 5 | 18 |
| 6 | 17 |

**a** Find the observed probability of rolling:

**i** an even number

**ii** an even number or a number greater than 4

'Even number or a number greater than 4' is an example of a **compound event**.

**iii** an even number less than or equal to 4.

**b** Calculate the probability of rolling a 2 or 3

**i** as an observed probability **ii** as an expected probability.

**c** If the die is rolled 100 times, what is the expected frequency of rolling a 2 or a 3?

How does this compare with James' observed frequency?

### Solution

**a i** Rolls of even numbers $= 19 + 12 + 17$ Frequencies of 2, 4, 6

$= 48$

Observed $P(\text{even}) = \frac{48}{100} = \frac{12}{25}$

**ii** Rolls of even numbers or numbers greater than 4 Frequencies of 2, 4, 6, 5

$= 19 + 12 + 17 + 18$

$= 66$

Observed $P(\text{even or} > 4) = \frac{66}{100} = \frac{33}{50}$

**iii** Rolls of even numbers less than or equal to 4 Frequencies of 2 and 4

$= 19 + 12$

$= 31$

Observed $P(\text{even and} \leq 4) = \frac{31}{100}$

**b i** Rolls of 2 or 3 $= 19 + 11$ Frequencies of 2 and 3

$= 30$

Observed $P(2 \text{ or } 3) = \frac{30}{100} = \frac{3}{10}$

**ii** Expected $P(2 \text{ or } 3) = \frac{2}{6} = \frac{1}{3}$

**c** Expected frequency of 2 or 3 $= \frac{1}{3} \times 100$ Probability × number of trials

$= 33.333...$

$\approx 33$

From the table, the observed frequency $= 19 + 11 = 30$, which is close to 33.

9.01

# Probability simulations

Sometimes it is impractical or impossible to run a probability experiment because it is expensive or time-consuming. In these cases, we can imitate or recreate a chance situation using a computer model or an object like a die or spinner to generate random events. This is called a **probability simulation**, which can be used to calculate observed probabilities and predict outcomes.

Spreadsheets, apps, cards, dice, spinners and coins can be used to simulate random events. Simulations allow many repetitions, often thousands of times, of chance experiments to be conducted. Because it can easily be repeated, an advantage of a simulation is that it can also be used to verify the theoretical probability of an event.

## Example 3

How likely is it for a family of 2 children to have both children of the same gender, that is, 2 boys or 2 girls? Design and conduct a probability simulation to determine a solution to this problem. Predict what the probability will be.

### Solution

As there are 2 possible outcomes for the gender of a child, it can be modelled by flipping a coin, for example, let heads represent girls (G) and tails represent boys (B).

To simulate a family with 2 children, flip 2 coins together (or flip one coin twice).

The possible outcomes for a family with 2 children are GG, GB, BG and BB.

**1** Copy the table below. Flip both coins after deciding which coin represents the first child. Alternatively search for an online coin flipping website or app or use the spreadsheet formula =INT(RAND()*2)+1) as shown in **Technology: Throwing a coin** after Exercise 9.01.

| Outcome | Tally | Frequency | Observed probability |
|---|---|---|---|
| GG | \|\|\| | | |
| GB | \| | | |
| BG | \|\| | | |
| BB | 𝍸 \|\|\| | | |
| | | Total: 30 | |

**2** Flip the 2 coins 30 times and complete the Tally column in the table.

**3** Complete the Frequency and Observed probability columns.

**4** Copy and complete the following to solve the problem.

$P(\text{same gender}) = P(\text{GG or BB})$

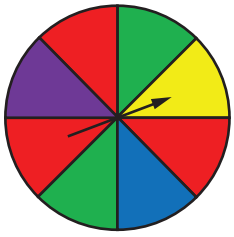
$= _____$

**5** Compare your results with other students in your class. Do you notice any similarities or differences? Why?

## EXERCISE 9.01 ANSWERS ON P. 615

### Observed probability U F P S R C

**1** Aashima spun this spinner 200 times and recorded the results. R C



| Event | Frequency |
|---|---|
| Red | 85 |
| Green | 42 |
| Blue | 28 |
| Yellow | 15 |
| Purple | 30 |

**a** Calculate, as a decimal, the observed probability that the arrow points to:

**i** red **ii** blue **iii** green.

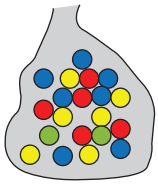
**b** Calculate, as a decimal, the theoretical probability that the arrow points to:

**i** red **ii** blue **iii** green.

**c** Are the observed and theoretical probabilities similar?

**d** For 200 spins, what is the expected frequency of red or purple based on the theoretical probability? How does this compare with the observed frequency?

**2** A bag contains 7 blue, 6 yellow, 5 red and 2 green marbles. Lamisa selects a marble at random, records its colour and then returns it to the bag. Lamisa repeats this process 100 times and the results are shown. R C



| Outcome | Frequency |
|---|---|
| Red | 20 |
| Blue | 38 |
| Yellow | 33 |
| Green | 9 |

**a** Find the observed probability of selecting a marble that is:

**i** red **ii** blue

**iii** yellow **iv** green.

**b** What is the expected probability of selecting a marble that is:

**i** red? **ii** blue?

**iii** yellow? **iv** green?

**c** Are the observed and expected probabilities similar?

**d** If the process is repeated 100 times, what is the expected frequency of a selecting a yellow or green marble? How does this compare with the observed frequency?

**3** A coin is tossed. R C

**a** What is the expected frequency of heads if the coin is tossed 100 times?

**b** Toss a coin 100 times, either by using a real coin or by computer or calculator simulation. Copy this table and record your results in it.

| Outcome | Frequency |
|---|---|
| Head | |
| Tail | |

**c** Calculate, as a decimal:

**i** the observed probability of tossing a head

**ii** the theoretical probability of tossing a tail.

**d** Are the observed and theoretical probabilities similar?

EXAMPLE 2

**4** A die was repeatedly rolled and the results are shown in the table. R C

| Outcome | Frequency |
|---|---|
| 1 | 95 |
| 2 | 119 |
| 3 | 108 |
| 4 | 87 |
| 5 | 78 |
| 6 | 113 |

**a** How many times was the die rolled?

**b** Find the observed probability (as a decimal) of rolling:

**i** an odd number

**ii** a number less than 4

**iii** a 2 or a 3

**iv** a number less than 4 or an even number.

**c** Find the expected probability (as a decimal) of rolling:

**i** an odd number

**ii** a number less than 4

**iii** a 2 or a 3

**iv** a number less than 4 or an even number.

**d** Compare the observed probabilities to the expected probabilities.

**5** Place 5 blue counters, 2 red counters and 3 yellow counters in a bag. Select a counter at random from the bag, note its colour and return the counter to the bag. Repeat this 60 times. Copy this table and record your results in it. R C

| Event | Tally | Frequency |
|---|---|---|
| Blue | | |
| Red | | |
| Yellow | | |

**a** What is the observed probability of selecting:

**i** a blue counter? **ii** a red counter?

**iii** a yellow counter? **iv** a red or blue counter?

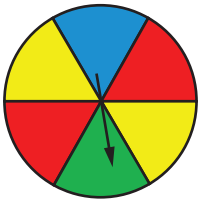
**b** What is the theoretical probability of selecting:

**i** a blue counter? **ii** a red counter?

**iii** a yellow counter? **iv** a red or blue counter?

**c** Are the observed probabilities similar to the theoretical probabilities?

**6** Tara spun this spinner 50 times and the results are shown. R C



| Event | Frequency |
|---|---|
| Red | 15 |
| Blue | 6 |
| Yellow | 24 |
| Green | 5 |

**a** What is the observed probability (as a decimal) of the arrow stopping on

**i** red? **ii** blue? **iii** yellow? **iv** green?

**b** What is the expected probability (as a decimal) of the arrow stopping on

**i** red? **ii** blue? **iii** yellow? **iv** green?

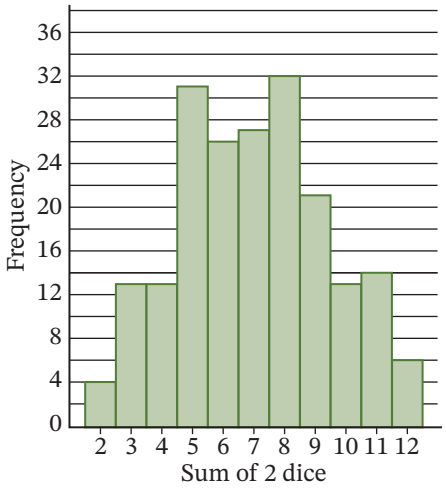
**c** Are the observed and expected probabilities similar?

**d** What is the expected frequency of the arrow stopping on a colour that is not yellow? How does this compare with Tara's observed frequency?

Foundation Standard Complex


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**7** 2 dice are rolled and the sum of the numbers rolled was recorded in the frequency histogram.




**a** How many times were the dice rolled?

**b** Based on these results, what is the observed probability of rolling a sum:

**i** of 2? **ii** of 7?

**iii** of 10? **iv** greater than 7?

**v** less than 7?

**vi** of 7 or 8?

**vii** that is even and greater than 6?

**8** Children at a shopping mall were asked how they travelled to school. R C

| Mode of transport | Frequency |
|---|---|
| Walk | 27 |
| Bus | 80 |
| Car | 62 |
| Train | 21 |
| Bicycle | 5 |
| Skateboard | 1 |
| Other | 4 |

**a** How many students were surveyed?

**b** Based on these results, find the probability that a student chosen at random will:

**i** walk to school

**ii** be driven to school

**iii** catch a bus to school

**iv** catch a train to school

**v** ride a skateboard to school.

**c** What mode of transport could 'Other' include?

**d** Survey 100 students at your school and make up a table showing the results. How do the results from your school compare with the results from the survey?

**9** A die is rolled 100 times. R C

EXAMPLE 3

| Outcome | Tally | Frequency |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |
| 6 | | |

**a** What is the expected probability of rolling a 6? (Express your answer as a fraction and as a decimal.)

**b** How many times would you expect a 6 to appear if the die was rolled 100 times?

**c** Simulate rolling a die 100 times and record your results in a table similar to the one shown.

**d** What is the relative frequency of rolling a 6? (Express your answer as a fraction and as a decimal.)

**e** How does the expected probability of rolling a 6 compare with the observed probability?

**10** A school trivia quiz awards winners by using a computer to select a random prize from the following: Rubik's cube, art supplies, lollies, chocolates, squishy toy and ice cream. Greta won 2 prizes. Design and conduct a probability experiment to determine the probability that both prizes were the same. PS R C

**11** A spinner for a game has 5 equal sectors with 2 of the sectors labelled WIN. Jayden spins it 3 times and wins the game if he spins WIN twice or more. Design a probability experiment to determine the probability of winning the game. An Internet search of 'adjustable probability spinners' could help you simulate this experiment. PS R C

**12** Two out of every 5 people who have a runny nose and a fever have tested positive for a virus. Design a probability simulation to predict the number of people infected by this virus from 1000 tests. PS R C

**13** Five out of every 7 people own a pet. Of these pet owners, 80% have a dog. Design a probability simulation to model this situation and calculate the percentage probability that a person selected at random owns a dog. PS R C

**14** 'If there are 10 horses in a race, the probability of each horse winning is 1 in 10'. Explain why this statement is not true. Give at least 2 reasons. R C



Mikhail Pogosov/Shutterstock.com

# Technology

## Throwing a coin

Worksheet
Coins probability

9.01

In this activity, a spreadsheet is used to simulate the throwing of a coin. The computer can quickly generate either a 1 (representing heads) or a 2 (tails).



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1. In cell A1, enter the label 'Tossing a coin'.
2. In cell A2, enter the formula **=INT(RAND()*2+1)**, press Enter and either 1 or 2 will appear in the cell.
3. Select cell A2 and **Fill Down** to cell A11 to simulate the tossing of a coin 10 times.
4. Copy the table and in the first blank row record your results for Trials 1–10 (the numbers of heads and tails in your first 10 'tosses').
5. Simulate another 10 tosses of the coin by making the spreadsheet generate another set of random numbers. Do this by placing the cursor between the tops of columns A and B (so that it turns into a 'double arrow') and clicking.
6. Enter the results for Trials 11–20 in your table.
7. Repeat 8 more times so that you have 100 tosses of the coin recorded in the table.
8. **a** For 100 tosses of a coin, what is the expected frequency of heads based on the theoretical probability?
   **b** Compare this with the actual frequency.
9. Calculate, as a decimal, the theoretical and observed probabilities of tossing heads.
10. Compare your results with those of other students in your class. Briefly explain any differences and why they may have occurred.
11. Combine the results of students in your class to calculate the observed probability of throwing heads, as a decimal. How does this compare with the theoretical probability?

| Trial | Number of heads | Number of tails |
|---|---|---|
| 1–10 | | |
| 11–20 | | |
| 21–30 | | |
| 31–40 | | |
| 41–50 | | |
| 51–60 | | |
| 61–70 | | |
| 71–80 | | |
| 81–90 | | |
| 91–100 | | |
| Total | | |

Worksheet
Dice probability

# Technology

## Rolling dice

In this activity, we will use a scientific calculator to simulate the rolling of a die.

**1** Copy the table shown.

| Outcome | Tally | Frequency | Observed probability |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |
| 6 | | | |

**2**

| Casio scientific calculator | Sharp scientific calculator |
|---|---|
| Enter the formula RanInt#(1,6) as shown: [ALPHA] [.] 1 [SHIFT] [)] 6 [)] [=] | [2nd F] 7 (RANDOM), select 1 R-DICE, [=] |

This generates a random number from 1 to 6. Press [=] more times to generate more random numbers. Alternatively, on a spreadsheet, type **=INT(RAND()*6+1)** into a cell and press the Enter key, then the [F9] key repeatedly for more random numbers.

**3** Run the simulation 50 times and record the results in the table.

**4** Calculate the theoretical probability for each number being tossed on the die.

**5** Compare the theoretical probability and observed probability for each number. Are there similarities or differences? How could the observed probabilities be more accurate? Explain.

**6** Copy this table for rolling 2 dice and finding their sum.

| Sum | Tally | Frequency | Observed probability |
|---|---|---|---|
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |
| 6 | | | |
| 7 | | | |
| 8 | | | |
| 9 | | | |
| 10 | | | |
| 11 | | | |
| 12 | | | |

**7** On a scientific calculator, rolling 2 dice is achieved by adding the result of rolling one die twice.

| Casio scientific calculator | Sharp scientific calculator |
|---|---|
| Enter the formula RanInt#(1,6) + RanInt#(1,6) as shown below:<br>ALPHA · 1 SHIFT ) 6 ) + <br>ALPHA · 1 SHIFT ) 6 ) = | Enter the formula R.DICE + R.DICE as shown below:<br>2nd F 7 (RANDOM), select 1 R-DICE,<br>+ 2nd F 7 (RANDOM), select 1 R-DICE, = |

This generates the sum of 2 random numbers from 1 to 6.

Press = more times to generate more sums.

**8** Run the simulation 50 times and record the results in the table.

**9** Which sum is the most likely to occur?

**10** Which sum is the least likely to occur? Why?

**11** Which sum is theoretically the:

**a** most likely? **b** least likely?

Why?

**12** Compare your answers to questions **9**, **10** and **11** and then compare them with the answers of other students in your class. Write a few sentences explaining your observations.

**13** If you ran 50 more simulations, would you expect your results to be similar or different to your first 50 results? Explain.



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# 9.02 Sets and Venn diagrams

PATH

**Videos**
Venn diagrams 1
Venn diagrams 2

**Puzzles**
Venn diagrams matching activity
And/or problems
Venn diagrams

## Sets

A **set** is a collection of distinct items, for example, a set of cricket equipment consists of a cricket bat, stumps, a cricket ball, helmet, pads, and so on. Distinct means 'different', so the elements in a set are all different.



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The members of a set are called its **elements** and they can be listed between **braces** (curly brackets), separated by commas. For example,

- the set of even numbers {2, 4, 6, 8, 10, ...}
- the set of odd numbers {1, 3, 5, 7, 9, ...}
- the set of prime numbers {2, 3, 5, 7, 11, 13, 17, 19, ...}
- the set of multiples of 7 {7, 14, 21, 28, 35, ...}

Consider the set of vowels in the alphabet, set $V = \{a, e, i, o, u\}$.

Then the **complementary set** to set $V$ is the set of letters in the alphabet that are *not* vowels.

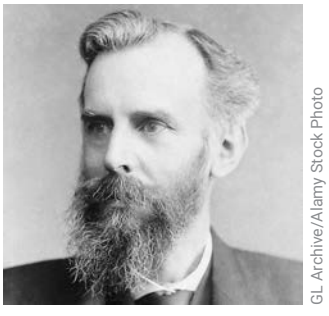
The complementary set to $V$ is written as $\overline{V}$ (or $V^C$ or $V'$).

$\overline{V} = \{b, c, d, f, g, h, j, k, l, m, n, p, q, r, s, t, v, w, x, y, z\}$ (the consonants)

$\overline{V}$ means 'not $V$' or the complement of $V$.

## Venn diagrams

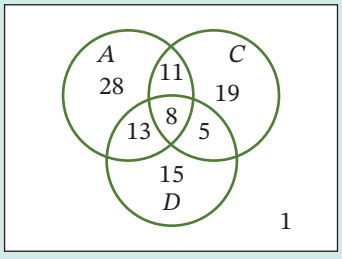
A **Venn diagram** is a diagram of circles (usually overlapping) that is used to represent sets. A rectangle represents the whole group, called the **universal set**, while the circles represent individual sets. Items common to 2 or more sets are placed in the intersection (overlapping region) of the circles. The Venn diagram was invented in 1880 by English mathematician and priest, John Venn (1834–1923).



GL Archive/Alamy Stock Photo

## Example 4

This Venn diagram shows the results of a survey on what type of movies – action ($A$), comedy ($C$) or drama ($D$) – students prefer to watch.




**a** How many students were surveyed?

**b** How many students preferred to watch 2 types of movies only?

**c** Calculate, as a decimal, the probability of selecting a student who prefers to watch:

**i** action movies only

**ii** action or comedy movies, but not dramas

**iii** action and drama movies

**iv** all types.

**d** A student is chosen from those who like action and comedy movies. What is the probability that they also like to watch drama movies?

**e** What is the probability of selecting a student who does not like watching any of the 3 movie types?

### Solution

**a** Number of students $= 28 + 11 + 8 + 13 + 5 + 19 + 15 + 1$

$= 100$

**b** 29 students preferred 2 types of movies only. $11 + 13 + 5 = 29$

**c** **i** Students preferring action movies only $= 28$ — The region of $A$ that doesn't overlap $C$ or $D$

$P(\text{action only}) = \frac{28}{100} = 0.28$

**ii** Students preferring action or comedy only — The regions of $A$ and $C$ that don't overlap with $D$

$= 28 + 19 + 11 = 58$

$P(\text{action or comedy only}) = \frac{58}{100} = 0.58$

'Action or comedy only' is an example of a **compound event**.

**iii** Students preferring action and drama $= 13 + 8$ — The regions where $A$ and $D$ intersect

$= 21$

$P(\text{action and drama}) = \frac{21}{100} = 0.21$

**iv** Students preferring all types $= 8$ — The region where the 3 circles intersect

$P(\text{all types}) = \frac{8}{100} = 0.08$

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**d** Students preferring action and comedy $= 11 + 8$

$= 19$

Students preferring action and comedy and drama $= 8$

$P(\text{drama if preferring action and comedy}) = \frac{8}{19}$

$\approx 0.42$

**e** There is one student who doesn't prefer action, comedy or drama.

$P(\text{not action, comedy or drama}) = \frac{1}{100}$

$= 0.01$

When we combine 2 or more simple events, we get a **compound event**. In the above example, 'action and drama' and 'all types' are compound events.

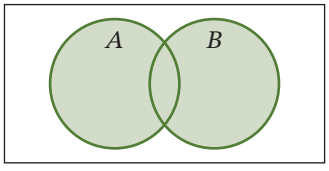
## Union and intersection

For 2 sets or events $A$ and $B$, the compound event '***A* or *B***' means to have $A$ or $B$ or both occurring. For example, 'to drive a car' **or** 'to ride a bus' means to drive a car only, or to ride a bus only, or to do both.

On a Venn diagram, this is represented by the region inside both sets $A$ and $B$.

In set notation, this is written $\boldsymbol{A \cup B}$ and is called the **union** of sets $A$ and $B$.

Union means to join, combine or unite.

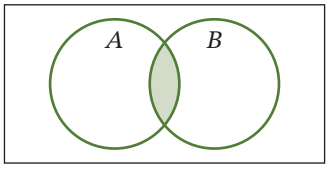



$A \cup B$ ('$A$ union $B$')

The compound event '***A* and *B***' means to have both $A$ and $B$ occurring together. For example, 'to drive a car' **and** 'to ride a bus' means to do both things.

On a Venn diagram, this is represented by the region where sets $A$ and $B$ overlap.

In set notation, this is written $\boldsymbol{A \cap B}$ and is called the **intersection** of sets $A$ and $B$.




$A \cap B$ ('$A$ intersection $B$')

Notice that '$A$ or $B$' actually **includes** '$A$ and $B$' so this is an example of an **inclusive 'or'**

If '$A$ or $B$' didn't include '$A$ and $B$', then that would be an **exclusive 'or'**.

## Union and intersection

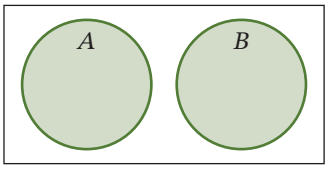
$A \cup B$ means '$A$ **union** $B$' ($A$ **or** $B$) and is the set of **all** elements in set $A$ and set $B$.

$A \cap B$ means '$A$ **intersection** $B$' ($A$ **and** $B$) and is the set of elements in **both** sets $A$ and $B$.

# Mutually exclusive events



If $A$ and $B$ are not overlapping, then they are **mutually exclusive** and on a Venn diagram they appear as 2 separate circles. For mutually exclusive sets or events, '***A* or *B***' means $A$ only or $B$ only but not both. For example, 'male' **or** 'female' means to be male, or female, but not both. In this case, '$A$ or $B$' **excludes** '$A$ and $B$' so this is an example of an **exclusive 'or'**.




**Mutually exclusive events:**
'$A$ or $B$' means $A$ or $B$ but not both

Also, $A \cap B = \{\ \}$ has no elements because mutually exclusive sets have nothing in common. $\{\ \}$ is called an **empty set**, also written $\varnothing$.

## Example 5

Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$ be the set of integers from 1 to 15.

**a** From this set, list the elements of each subset.

A **subset** is a set that is part of a larger set.

**i** $A = \{\text{odd numbers}\}$ **ii** $B = \{\text{even numbers}\}$ **iii** $C = \{\text{multiples of 5}\}$

**iv** $D = \{\text{prime numbers}\}$ **v** $E = \{\text{factors of 14}\}$

**b** List the elements of each set described.

**i** $A \cup C$ **ii** $C \cup D$ **iii** $B \cap C$ **iv** $A \cap D$

**v** $\overline{D}$ **vi** $C \cup \overline{E}$ **vii** $C \cap \overline{E}$ **viii** $\overline{C} \cap \overline{E}$

### Solution

**a** **i** $A = \{1, 3, 5, 7, 9, 11, 13, 15\}$ odd numbers from 1 to 15

**ii** $B = \{2, 4, 6, 8, 10, 12, 14\}$ even numbers from 1 to 15

**iii** $C = \{5, 10, 15\}$ multiples of 5 numbers from 1 to 15

**iv** $D = \{2, 3, 5, 7, 11, 13\}$ prime numbers from 1 to 15

**v** $E = \{1, 2, 7, 14\}$ factors of 14 from 1 to 15

**b** Using the answers from part **a**:

**i** $A \cup C = \{1, 3, 5, 7, 9, 10, 11, 13, 15\}$ $A$ union $C$

**ii** $C \cup D = \{2, 3, 5, 7, 10, 11, 13, 15\}$ $C$ union $D$

**iii** $B \cap C = \{10\}$ $B$ intersection $C$

**iv** $A \cap D = \{3, 5, 7, 11, 13\}$ $A$ intersection $D$

**v** $\overline{D} = \{1, 4, 6, 8, 9, 10, 12, 14, 15\}$ The complement of $D$ (not $D$)

**vi** $\overline{E} = \{3, 4, 5, 6, 8, 9, 10, 11, 12, 13, 15\}$ Not $E$

$C \cup \overline{E} = \{3, 4, 5, 6, 8, 9, 10, 11, 12, 13, 15\}$ $C$ union $\overline{E}$

**vii** $C \cap \overline{E} = \{5, 10, 15\}$ $C$ intersection $\overline{E}$

**viii** $\overline{C} = \{1, 2, 3, 4, 6, 7, 8, 9, 11, 12, 13, 14\}$ Not $C$

$\overline{C} \cap \overline{E} = \{3, 4, 6, 8, 9, 11, 12, 13\}$. Intersection of not $C$ and not $E$

PATH

## Example 6

A survey of 110 students at Hamper Valley College showed that 34 students study Art, 65 students study Chemistry, and 23 students study both Chemistry and Art.

**a** Represent this information on a Venn diagram.

**b** How many students study Art and Chemistry, but not both?

**c** What is the probability of randomly selecting a student from this group who takes:

- **i** Chemistry?
- **ii** Art and Chemistry?
- **iii** Art or Chemistry?
- **iv** neither Art nor Chemistry?



Alamy Stock Photo/Ian Shaw

### Solution

**a** S = students surveyed

A = students doing Art

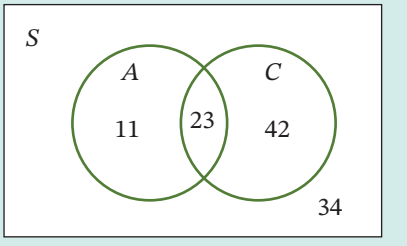
C = students doing Chemistry

There are 23 students who take both Art and Chemistry.

∴ Students doing Art only = 34 − 23 = 11

∴ Students doing Chemistry only = 65 − 23 = 42

∴ Students who take neither Art nor Chemistry = 110 − 11 − 42 − 23 = 34




**b** Number of students studying Art only or Chemistry only = 11 + 42 = 53

'Art or Chemistry only' is a compound event and an example of 'exclusive or'.

**c** **i** 65 students take Chemistry.

$P(\text{Chemistry}) = \frac{65}{110} = \frac{13}{22}$

**ii** $P(\text{Art and Chemistry}) = \frac{23}{110}$ $P(A \cap C)$

**iii** Number of students who take Art or Chemistry = 11 + 23 + 42 = 76

$P(\text{Art or Chemistry}) = \frac{76}{110} = \frac{38}{55}$ $P(A \cup C)$

**iv** $P(\text{neither Art nor Chemistry}) = \frac{34}{110} = \frac{17}{55}$ $P(\overline{A \cup C})$


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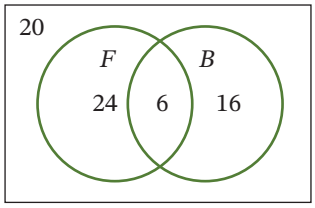
EXAMPLE 4

9.02

**EXERCISE 9.02** ANSWERS ON P. 616

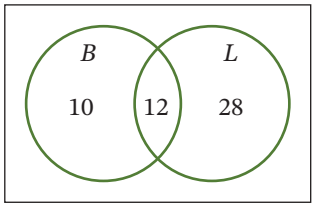
## Sets and Venn diagrams U F R C

**1** This Venn diagram shows the number of competitors who swam in freestyle ($F$) and butterfly ($B$) events at the school swimming carnival. How many students swam in freestyle or butterfly, but not both?
Select the correct answer **A**, **B**, **C** or **D**.




**A** 28 **B** 34 **C** 40 **D** 60

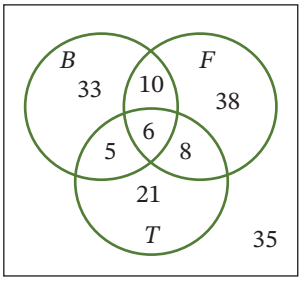
**2** 50 people were asked whether they had breakfast ($B$) or lunch ($L$) today. The results are shown in the Venn diagram. C




**a** What is the probability of selecting a person from this group who had:
- **i** breakfast?
- **ii** lunch?
- **iii** breakfast but not lunch?
- **iv** breakfast and lunch?
- **v** breakfast or lunch only?

**b** Of the people who had lunch, find the probability that a person also had breakfast.

**3** The Venn diagram shows the number of Year 10 students who play basketball ($B$), touch football ($F$) or tennis ($T$). C

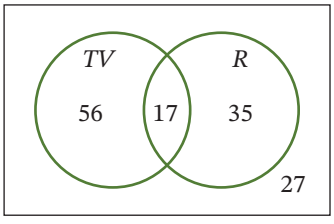



**a** How many students are in Year 10?

**b** Find the probability of selecting a student who plays:
- **i** basketball only
- **ii** tennis only
- **iii** touch football and tennis
- **iv** touch football or tennis
- **v** basketball but not touch football
- **vi** all 3 sports.

**c** Of the students that play touch football, find the probability of selecting a student who also plays tennis.

**4** This Venn diagram shows the results of a survey asking people whether they watch TV or read ($R$) at home. R




**a** How many people were surveyed?

**b** Find the probability of selecting a person who watches TV only.

**c** What is the probability of selecting a person who doesn't watch TV or read?

**d** Of the people who read, find the probability that they also watch TV.

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EXAMPLE 5

**5** Let $K = \{5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22\}$ be the set of integers from 5 to 22. R C

**a** From this set, list the elements of each subset.

**i** $M = \{\text{odd numbers}\}$ **ii** $N = \{\text{even numbers}\}$ **iii** $P = \{\text{multiples of 4}\}$

**iv** $Q = \{\text{square numbers}\}$ **v** $R = \{\text{prime numbers}\}$

**b** List the elements of each set described.

**i** $M \cup N$ **ii** $M \cup P$ **iii** $N \cap P$

**iv** $M \cap Q$ **v** $Q \cap R$

**6** Let $X = \{\text{vowels}\}$ and $Y = \{\text{letters from the word MUSICIAN}\}$. R C

**a** Represent sets $X$ and $Y$ on a Venn diagram.

**b** List the elements of the set $X \cap Y$.

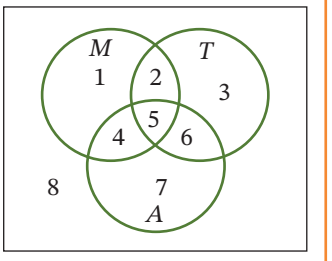
**c** List the elements of the set $X' \cap Y$.

**7** Consider this Venn diagram that has its regions numbered 1 to 8. R C

What regions are included in each set shown below?

**a** $M \cap A$ **b** $A \cup T$ **c** $\overline{M}$

**d** $(T \cap M) \cup A$ **e** $A \cap (T \cup M)$ **f** $\overline{T} \cap A$




**8** Let $A = \{31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46\}$ with subsets $B = \{\text{multiples of 4}\}$ and $C = \{\text{multiples of 8}\}$. List the elements of each set. R C

**a** $B \cup C$ **b** $B \cap C$ **c** $B \cup \overline{C}$ **d** $\overline{B} \cap C$

**e** $\overline{B} \cup C$ **f** $B \cap \overline{C}$ **g** $\overline{B} \cup \overline{C}$ **h** $\overline{B} \cap \overline{C}$

EXAMPLE 6

**9** Of the 54 Year 10 Music students, 23 students sing ($S$), 43 students play a musical instrument ($P$) and 12 students sing and play a musical instrument. R C

**a** Show this information on a Venn diagram.

**b** Find the probability of selecting a Music student who:

**i** sings or plays an instrument

**ii** sings only

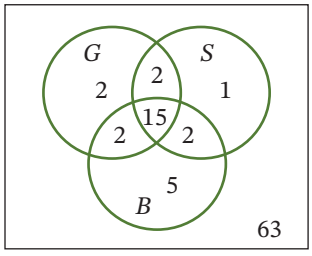
**iii** plays a musical instrument only

**iv** sings or plays an instrument, but not both.

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**10** This Venn diagram shows the number of countries that won medals at the Winter Olympic Games in Pyeongchang, South Korea in 2018. R




**a** How many countries competed at these games?

**b** What is the probability of randomly selecting a country that:

**i** won only gold medals?

**ii** won gold, silver and bronze medals?

**iii** won gold or silver medals, but not bronze?

**iv** did not win a gold or silver medal?

**c** Of the countries that won medals, what is the probability of selecting a country that

**i** won gold medals?

**ii** won bronze, but no gold or silver?

**11** At Riverside College, Year 10 students are asked what language they are studying. 64 students take French ($F$), 47 students take Japanese ($J$), 15 students take both French and Japanese, and 27 do not study a language. R C

**a** How many students are in Year 10?

**b** Show the information on a Venn diagram.

**c** How many students studied only one language?

**d** Find the probability of selecting a Year 10 student at random who studies:

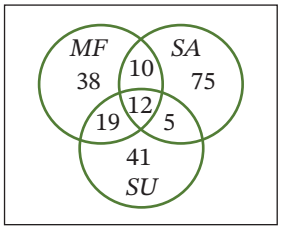
**i** French but not Japanese

**ii** Japanese but not French

**iii** no languages

**iv** only one language.

**12** People were surveyed on the day they preferred to shop: Monday to Friday ($MF$), Saturday ($SA$) or Sunday ($SU$). The results are shown in the Venn diagram. R C




**a** How many people were surveyed?

**b** What is the probability of selecting a person who prefers to shop on:

**i** Monday to Friday?

**ii** Saturday?

**iii** Sunday?

**iv** on the weekend only?

**v** on Saturday or Sunday?

**vi** any day (has no preference)?

**c** Find the probability of selecting a person who only prefers Saturday or Sunday, but not both.

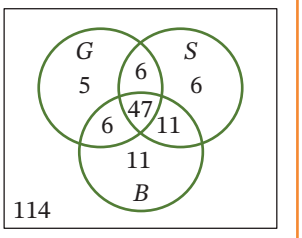
**d** Is it necessary to include the rectangle in this Venn diagram? Give reasons.



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**13** This Venn diagram shows the number of countries that won gold, silver, bronze or no medals at the 2021 Olympic Games in Tokyo, Japan.




**a** Find the total number of countries that competed at these games.

**b** What is the probability of randomly selecting a country that:

**i** won a silver medal only?

**ii** won one medal only?

**iii** won at least 2 medals?

**iv** won at most one medal only?

**c** Out of the countries that won gold medals, find the probability of selecting a country that:

**i** won gold and silver, but not bronze

**ii** won gold, silver and bronze.

**14** A travel agent surveyed 100 people to find out how many of them had visited Melbourne, $M$, and Brisbane, $B$. Of this sample, 31 people had visited Melbourne, 26 people had visited Brisbane and 12 people had visited both Melbourne and Brisbane. Draw a Venn diagram to represent these values and count the number of people who have visited: R C

**a** Melbourne or Brisbane, $M \cup B$

**b** Brisbane but not Melbourne, $B \cup M'$

**c** Melbourne or Brisbane, but not both, $(M \cup B) \cap (\overline{M \cap B})$

**d** neither Melbourne nor Brisbane, $(\overline{M \cup B})$.

**15** 24 people go on holidays. 15 go swimming, $S$, 12 go fishing, $F$, and 6 do neither activity. Draw a Venn diagram, find the number of elements in $S \cap F$ and describe what the people in this region do. R C

**16** At Nelson Cove Senior College, students can study any of the 3 languages, French, $F$, Japanese, $J$, and Spanish, $S$. Of the 35 Year 12 Languages students, 21 study French, 18 study Japanese and 13 study Spanish. Of these, 11 students study both French and Japanese, 7 study both French and Spanish, 6 study both Japanese and Spanish, while one student studies all 3 languages. R C

**a** Represent the given information on a Venn diagram.

**b** How many students do not study any of the languages, $\overline{F} \cap \overline{J} \cap \overline{S}$?

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# Two-way tables

9.03

A **two-way table** is another way of grouping items into overlapping categories, especially when there are many overlaps that cannot easily be represented by a Venn diagram.

## Example 7

Year 11 students at Southbank College were surveyed on whether they had part-time jobs.

| | Part-time work | No part-time work |
|---|---|---|
| Male | 43 | 27 |
| Female | 35 | 31 |

**a** How many students are in Year 11 at Southbank College?

**b** How many students had part-time work?

**c** How many male students were in Year 11?

**d** What is the probability of selecting a student at random that:

**i** works part-time? **ii** is female and works part-time?

**iii** is male and doesn't work? **iv** doesn't work?

**e** What is the probability of selecting a student working part-time given that:

**i** the student is male? **ii** the student is female?

### Solution

**a** Number of Year 11 students $= 43 + 27 + 35 + 31 = 136$

**b** Students with part-time work $= 43 + 35 = 78$

**c** Male students in Year 11 $= 43 + 27 = 70$

**d** **i** $P(\text{student works part-time}) = \frac{78}{136} = \frac{39}{68}$

**ii** There are 35 female students who work part-time.

$P(\text{female and part-time}) = \frac{35}{136}$

'Female and works part-time' is a compound event.

**iii** There are 27 males who don't work.

$P(\text{male and not working}) = \frac{27}{136}$

**iv** Number of students not working $= 27 + 31 = 58$

$P(\text{not working}) = \frac{58}{136} = \frac{29}{68}$

**e** **i** There are 70 male students and 43 of them work part-time.

$P(\text{working part-time given that student is male}) = \frac{43}{70}$

**ii** There are 66 female students and 35 of them work part-time.

$P(\text{working part-time given that student is female}) = \frac{35}{66}$

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9.03

**Videos**
Two-way tables 1
Two-way tables 2

**Worksheets**
Two-way tables
Two-way probability tables

**Puzzle**
Combined events: Two-way tables

PATH

## EXERCISE 9.03 ANSWERS ON P. 617

### Two-way tables U F R C

EXAMPLE 7

**1** People attending the Staying Alive Fitness Centre early on a Saturday morning either went swimming or did a workout in the gym. The numbers are shown in the table. R C

| | Swimming | Gym |
|---|---|---|
| Male | 32 | 53 |
| Female | 24 | 41 |

**a** How many people went to the fitness centre?

**b** Find the probability that a person selected at random:

**i** was female and went swimming

**ii** was male and did a workout in the gym

**iii** went swimming.

**c** Find the percentage (to the nearest whole number) of females who did a workout in the gym.

**2** Year 10 students at Nusentry High School were given a choice of 2 activities on a wet sports afternoon: bowling or indoor soccer. Their selections are shown in the table. R C

| | Bowling | Indoor soccer |
|---|---|---|
| Boys | 25 | 48 |
| Girls | 43 | 12 |

**a** How many students are in Year 10?

**b** How many students:

**i** went bowling? **ii** played indoor soccer?

**c** What is the probability of randomly selecting a student who went bowling?

**d** What is the probability of randomly selecting a girl who played indoor soccer?

**3** The composition of the Legislative Assembly (the lower house) in the State Parliament is shown in the table. R C

| | Liberal/Nationals | Labor | Greens/Other |
|---|---|---|---|
| Male | 41 | 20 | 5 |
| Female | 11 | 14 | 6 |

The Liberal/Nationals are in government and the others are in opposition.

**a** How many members of parliament (MPs) are there in the Legislative Assembly?

**b** Find the percentage probability of randomly selecting an MP who is:

**i** female **ii** male and in the Opposition **iii** Greens/Other.

**c** What percentage probability of:

**i** Government MPs are female? **ii** Opposition MPs are female?

**d** Compare your answers to part **c** and comment on the differences between the 2 results.

**4** People were asked to name their favourite takeaway food. The results are shown in the table. R C

| | Pizza | Hamburger | Fish and Chips |
|---|---|---|---|
| Men | 22 | 35 | 18 |
| Women | 43 | 26 | 6 |

**a** How many people were surveyed?

**b** Find the probability (as a decimal) that a person selected at random:

**i** is male

**ii** is female and likes fish and chips

**iii** likes pizza

**iv** is male and likes hamburgers.

**c** If a male is selected at random, what is the probability that his favourite takeaway food is pizza?

**5** Year 7 students were asked about their favourite drink. The results are in the table. R C

| | Boys | Girls |
|---|---|---|
| Water | 21 | 35 |
| Milk | 11 | 12 |
| Juice | 15 | 17 |
| Soft drink | 31 | 18 |

**a** How many students were in Year 7?

**b** What is the probability of randomly selecting a student that:

**i** prefers water?

**ii** is a boy and likes milk?

**iii** is a girl and likes soft drinks?

**c** What is the probability that if a girl is randomly selected, she prefers water?

**6** A survey looked at whether people ate breakfast and whether they exercised regularly. R C

| | Exercise | No exercise |
|---|---|---|
| Ate breakfast | 72 | 27 |
| Did not eat breakfast | 38 | 63 |



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**a** How many people were surveyed?

**b** What percentage of people exercised?

**c** Find the percentage probability of picking a person at random who:

**i** eats breakfast

**ii** does not exercise regularly

**iii** eats breakfast and exercises regularly

**iv** does not eat breakfast and does not exercise.

**d** Of the people who exercise regularly, what is the probability of picking someone who eats breakfast?



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PATH

**7** Students at Mt Badger College were asked to indicate their preference for dark or milk chocolate in a survey. R C

| | | Milk chocolate | |
|---|---|---|---|
| | | Like | Dislike |
| Dark chocolate | Like | 545 | 134 |
| | Dislike | 157 | 42 |

**a** How many students attended the college?

**b** What is the probability (correct to 3 decimal places) of selecting a student at random who:

**i** likes dark chocolate?

**ii** likes both milk chocolate and dark chocolate?

**iii** likes dark chocolate, but dislikes milk chocolate?

**iv** dislikes both dark and milk chocolate?

# 9.04 Tree diagrams

**Video** Tables and tree diagrams

**Worksheet** Tree diagrams

**Puzzle** Combined events: Tree diagrams

A **two-** or **three-stage experiment** is a **chance experiment** that has 2 or 3 parts or stages, for example:

- rolling 2 or 3 dice
- drawing 2 or 3 prizes in a raffle
- observing the weather each day over a weekend or a long weekend
- throwing 2 or 3 coins together.

A **tree diagram** lists all the possible outcomes of each stage. Branches stretch out to show the possible pathways of outcomes at each stage. An outcomes column at the end of the diagram lists the sample space.

The sample space for two-stage experiments can be displayed using lists, tables or tree diagrams, but the sample space for three-stage experiments is best displayed using a tree diagram.

## Example 8

A coin is tossed and a die is rolled.

**a** Use a table to display the sample space.

**b** Find the probability of rolling:

**i** a tail and a 3

**ii** a head and an even number.

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## Solution

**a** The sample space of a coin is a head (H) and a tail (T).

The sample space for a die is 1, 2, 3, 4, 5 and 6.

The sample space of tossing a coin and rolling a die is shown in the table below.

| | | Die | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 | 6 |
| Coin | H | H1 | H2 | H3 | H4 | H5 | H6 |
| | T | T1 | T2 | T3 | T4 | T5 | T6 |

Using a table ensures that all outcomes are counted

**b i** There are 12 outcomes in the sample space.

$\therefore P(\text{a tail and a 3}) = P(\text{T3}) = \frac{1}{12}$

'Tail and 3' is a compound event.

**ii** There are 3 outcomes that make up the event a head and an even number: H2, H4, H6.

$\therefore P(\text{a head and an even number}) = \frac{3}{12}$

$= \frac{1}{4}$

## Example 9

2 coins are tossed.

**a** Use a tree diagram to list the sample space.

**b** Find the probability of tossing:

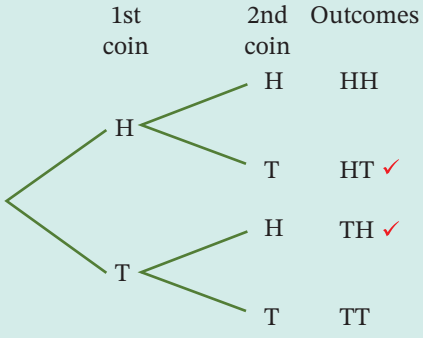
**i** 2 heads

**ii** a head and a tail (in any order).

## Solution

**a** There are 2 outcomes for the first coin, followed by 2 outcomes for the second coin. There are $2 \times 2 = 4$ possible outcomes.

Using a tree diagram ensures that all outcomes are counted.




**b i** There is one outcome out of a possible 4 for 2 heads.

$\therefore P(\text{2heads}) = \frac{1}{4}$

**ii** There are 2 outcomes for a head and a tail (✓ on the tree diagram).

$\therefore P(\text{a head and a tail}) = \frac{2}{4}$

$= \frac{1}{2}$

9.04

**Videos**
Tree diagrams 3
Tree diagrams 1
Tree diagrams 2

## EXERCISE 9.04 ANSWERS ON P. 617

## Tree diagrams U F PS R C

EXAMPLE 8

**1** A boy and a girl are to be chosen from this shortlist to represent the school at a conference:

**Boys:** Ben, Christian, Ewan, William.

**Girls:** Becky, Cassandra, Emily, Marianne, Roz, Siana. R C

**a** List all the possible pairs of a boy and a girl.

**b** Find the probability of selecting:

**i** Christian and Roz

**ii** a boy and a girl whose names begin with a B or a C

**iii** a pair that includes Emily.

**2** 2 coins are tossed together. R C

**a** Copy and complete the table to find all the outcomes in the sample space.

| | | 1st coin | |
|---|---|---|---|
| | | H | T |
| 2nd coin | H | | |
| | T | | |

**b** What is the probability of tossing:

**i** 2 tails? **ii** a head and a tail?

**iii** at least one head?

**3** 2 dice are rolled.

**a** Copy and complete this table to list the sample space. PS R C

| | | 1st die | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 | 6 |
| 2nd die | 1 | 1, 1 | 2, 1 | | | | |
| | 2 | | | | | | |
| | 3 | | | | | | |
| | 4 | | | 3, 4 | | | |
| | 5 | | | | | | |
| | 6 | | | | | | |

**b** How many possible outcomes are there?

**c** Find the probability of rolling:

**i** two 6s **ii** at least one 6

**iii** doubles **iv** 2 even numbers

**v** at least one 2 **vi** 2 numbers greater than 3

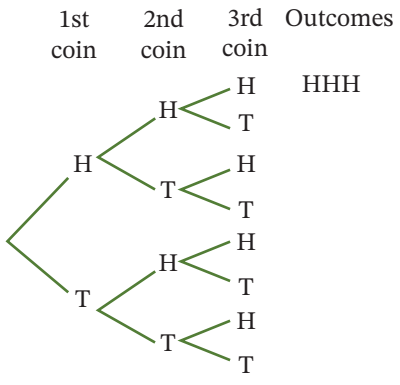
**vii** one odd and one even number

**viii** 2 numbers where the first number is greater than the second number.

EXAMPLE 9

9.04

**4** 3 coins are tossed. PS R C




**a** Copy and complete the tree diagram.

**b** How many outcomes are there in the sample space?

**c** Use the tree diagram to find:

**i** $P$(3 heads) **ii** $P$(2 heads)

**iii** $P$(3 tails)

**iv** $P$(head, then tail and then head)

**v** $P$(2 heads or 3 heads).

**d** Find the probability of tossing:

**i** at least 1 tail **ii** at most 2 tails.

**e** If 3 coins are tossed 200 times, find the expected frequency of:

**i** tossing 2 heads **ii** tossing no tails.

**5** Use a tree diagram to display all possible outcomes when a coin and die are tossed together. R C

**6** A 6-sided die and a 4-sided die (numbered 1, 2, 3 and 4) are rolled together. PS R C

**a** Construct a table to list the outcomes in the sample space.

**b** How many outcomes are in the sample space?

**c** Find the probability of rolling:

**i** doubles

**ii** 2 even numbers

**iii** one even and one odd number

**iv** a pair of numbers that are both less than 4

**v** a pair of numbers that are both greater than 4.

**7** 2 dice are rolled and the sum of the 2 numbers is calculated. PS R C

**a** Copy and complete this table to show all possible sums.

| | | 1st die | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 | 6 |
| 2nd die | 1 | 2 | | | | | |
| | 2 | | | | | | |
| | 3 | | | | 7 | | |
| | 4 | | | | | | |
| | 5 | | | | | | |
| | 6 | | | | | 11 | |

**b** Find the of probability of rolling a sum:

**i** of 5 **ii** of 12 **iii** of 7

**iv** that is even **v** less than 2 **vi** more than 7

**vii** at least 7 **viii** between 4 and 8.

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**8** 4 coins are tossed. PS R C

**a** Use a tree diagram to list the sample space.

**b** Find the probability of tossing:

**i** 4 heads **ii** 1 head **iii** 2 tails

**iv** at least 1 tail **v** 2 heads and then 2 tails **vi** not more than 1 tail.

**c** If 4 coins are tossed 1000 times, find the expected frequency of having:

**i** 4 heads **ii** 2 heads and 2 tails **iii** at least one tail.

**9** The weather on a long weekend will either be fine or rain each day, with each outcome being equally likely. PS R C

**a** Draw a tree diagram to show the possible outcomes for Saturday, Sunday and Monday.

**b** What is the probability that:

**i** it rains on all 3 days?

**ii** it is fine on 2 of the 3 days?

**iii** it is fine on Saturday and Sunday, but rains on Monday?

**iv** it rains on at least one day of the long weekend?

Quiz
Mental skills 9

# Mental skills 9: Maths without calculators ANSWERS ON P. 618

## Percentage increase and decrease

The fraction equivalents of commonly-used percentages can help us when we need to increase or decrease a number by a percentage.

| Percentage | 1% | 5% | 10% | $12\frac{1}{2}\%$ | 20% | 25% | $33\frac{1}{3}\%$ | 50% |
|---|---|---|---|---|---|---|---|---|
| Fraction | $\frac{1}{100}$ | $\frac{1}{20}$ | $\frac{1}{10}$ | $\frac{1}{8}$ | $\frac{1}{5}$ | $\frac{1}{4}$ | $\frac{1}{3}$ | $\frac{1}{2}$ |

**1** Study each example.

**a** Increase 360 by 25%.

$25\% \text{ of } 360 = \frac{1}{4} \times 360$

$= 360 \div 4$

$= 90$

$360 + 90 = 450$

**b** Increase \$80 by 5%.

$5\% \text{ of } \$80 = \frac{1}{20} \times \$80$

$= \$80 \div 20$

$= \$4$

$\$80 + \$4 = \$84$

or $10\% \text{ of } \$80 = \$8$

$\therefore 5\% \text{ of } \$80 = \$8 \div 2$

$= \$4$

**2** Now increase:

**a** \$340 by 20% **b** 66 by 50% **c** 150 by $33\frac{1}{3}\%$ **d** \$400 by 1%

**e** 640 by 5% **f** \$72 by $12\frac{1}{2}\%$ **g** \$470 by 10% **h** 180 by 25%

**i** 420 by $33\frac{1}{3}\%$ **j** \$80 by 5% **k** \$280 by 25% **l** 70 by 20%

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**3** Study each example.

**a** Decrease \$225 by $33\frac{1}{3}\%$

$33\frac{1}{3}\%$ of $\$225 = \frac{1}{3} \times \$225$

$= \$225 \div 3$

$= \$75$

$\$225 - \$75 = \$150$

**b** Decrease \$70 by 15%

10% of $\$70 = \frac{1}{10} \times \$70 = \$7$

$\therefore$ 5% of $\$70 = \frac{1}{2} \times \$7 = \$3.50$

$\therefore$ 15% of $\$70 = (10\% \times \$70) + (5\% \times \$70)$

$= \$7 + \$3.50$

$= \$10.50$

**4** Now decrease:

**a** 440 by 25%
**b** \$300 by 20%
**c** 2400 by $33\frac{1}{3}\%$
**d** \$500 by 15%
**e** \$250 by 10%
**f** \$120 by 50%
**g** \$72 by $12\frac{1}{2}\%$
**h** 80 by 5%
**i** \$85 by 20%
**j** \$3800 by 1%
**k** \$440 by 15%
**l** \$150 by $33\frac{1}{3}\%$

## Investigation

### The birth month paradox

A **paradox** is a statement or proposition that seems impossible but is actually true.

**1** Copy this table.

| Group | Outcome (Y or N) |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |

**2** Randomly select a group of 5 people and ask them what month they were born in. If 2 or more people have the same birth month, record a Y in the table for Group 1, otherwise write N.

**3** Repeat this process 4 more times, recording your results in the table.

**4** Combine your results with those of 6 other students so that you have the outcomes for 30 groups.

**5** What fraction of the groups had repeated birth months?

**6** Collect the results of another group of 6 students. What fraction of the groups had a repeated birth month?

**7** The birth month paradox is that in any randomly selected group of 5 people, the probability that at least 2 people have the same birth month is greater than 0.5. Have your results shown this to be true?

**8** Can you show the following?

**a** For every 23 people selected at random, the probability that at least 2 people will share the same birthday is 50%.

**b** If 30 people are selected at random, this probability is 70%.

**c** If 50 people are selected at random, this probability is 97%.

# 9.05 Selecting with and without replacement

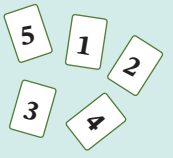
Worksheet
Multi-step experiments

Video
Selecting with and without replacement

In two- and three-stage experiments where an item is selected repeatedly, the outcome of the second or third stage may be affected by the outcome of the previous stage. This depends on whether each selected item is **returned** to the set of items before the next item is selected. If it is, then this is called **selecting 'with replacement'**. If it isn't, then it is called **selecting 'without replacement'**.

## Example 10

2 cards are selected at random from a set of cards numbered 1 to 5, to form a 2-digit number.




**a** Make a list of all possible outcomes if the cards are drawn:
- **i** with replacement
- **ii** without replacement.

**b** If the first card is replaced before the second card is drawn, find the probability that the number formed is:
- **i** even
- **ii** greater than 30
- **iii** divisible by 5.

**c** If the first card is not replaced, find the probability that the number formed is:
- **i** even
- **ii** odd
- **iii** less than 20.

### Solution

**a i** The possible outcomes, **with replacement**, are:

| | | 1st digit | | | | |
|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 |
| 2nd digit | 1 | 11 | 21 | 31 | 41 | 51 |
| | 2 | 12 | 22 | 32 | 42 | 52 |
| | 3 | 13 | 23 | 33 | 43 | 53 |
| | 4 | 14 | 24 | 34 | 44 | 54 |
| | 5 | 15 | 25 | 35 | 45 | 55 |

There are $5 \times 5 = 25$ different outcomes possible.

**ii** The possible outcomes, **without replacement**, are:

| | | 1st digit | | | | |
|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 |
| 2nd digit | 1 | – | 21 | 31 | 41 | 51 |
| | 2 | 12 | – | 32 | 42 | 52 |
| | 3 | 13 | 23 | – | 43 | 53 |
| | 4 | 14 | 24 | 34 | – | 54 |
| | 5 | 15 | 25 | 35 | 45 | – |

There are fewer outcomes for 'without replacement' because numbers with repeated digits such as 11 and 44 are not allowed.

There are $5 \times 4 = 20$ different outcomes possible.

**b i** With replacement, there are 10 even numbers.

$P(\text{even number}) = \frac{10}{25} = \frac{2}{5}$


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**ii** There are 15 numbers greater than 30.

$P(\text{number} > 30) = \frac{15}{25} = \frac{3}{5}$

**iii** There are 5 numbers divisible by 5. 15, 25, 35, 45 and 55

$P(\text{number divisible by 5}) = \frac{5}{25} = \frac{1}{5}$

**c i** Without replacement, there are 8 even numbers.

$P(\text{even}) = \frac{8}{20} = \frac{2}{5}$

**ii** There are 12 odd numbers.

$P(\text{odd}) = \frac{12}{20} = \frac{3}{5}$

**iii** There are 4 numbers less than 20.

$P(\text{number} < 20) = \frac{4}{20} = \frac{1}{5}$



## Probability tree diagrams

For more complex probability problems, the number of possible outcomes can be quite large and the tree diagram very long and detailed. In these cases, it is often possible to construct a simplified tree diagram that has probability values listed on its branches. This smaller 'trimmed' tree diagram is sometimes called a **probability tree diagram**.

### Example 11

A bag contains 3 red lollies, 5 blue lollies and 4 green lollies. Two lollies are selected at random from the bag without replacement. Calculate the probability of selecting:

**a** red lolly and then a blue lolly **b** 2 lollies the same colour.

#### Solution

First, write down the numbers of lollies and count the total.

3 R

5 B

4 G 12 lollies

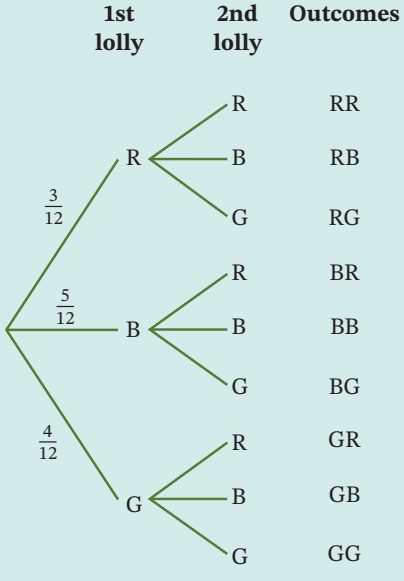
The tree diagram will be 2 stages deep as there are 2 selections.

For the 1st lolly, draw 3 branches for the 3 colours: red (R), blue (B), green (G).

For the 2nd lolly, also draw 3 branches for the 3 colours.

List the $3 \times 3 = 9$ outcomes from RR down to GG.

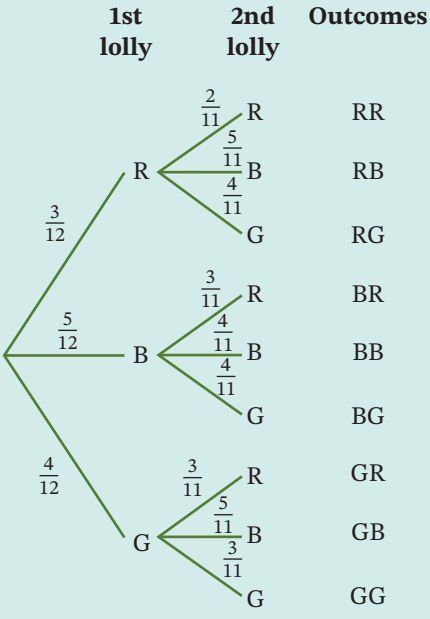
For the 1st lolly, the possible outcomes are red, blue or green. Write the appropriate probability values out of 12 for each colour: $\frac{3}{12}, \frac{5}{12}, \frac{4}{12}$.



For the 2nd lolly, there are now 11 lollies left. Write the appropriate probability values out of 11 for each colour. Make sure that the probabilities consider the first counter taken out.

If the first counter was red, then there are only 2 red lollies left in the bag, so the probability of another red is $\frac{2}{11}$.

Hint: The fractions in each group of joined branches should add up to 1, for example, $\frac{2}{11}+\frac{5}{11}+\frac{4}{11}=\frac{11}{11}=1$.




**a** To find the probability of red and then blue, follow the branches that give RB and multiply their probabilities.

$P(\text{RB}) = \frac{3}{12}\times\frac{5}{11}$ 'AND' means we multiply the probabilities

$= \frac{15}{132}$

$= \frac{5}{44}$

**b** To find the probability of 2 same colours, look for RR or BB or GG.

$P(\text{same colour}) = \text{P(RR or BB or GG)}$ 'OR' means we add the probabilities

$= \text{P(RR)} + \text{P(BB)} + \text{P(GG)}$

$= \frac{3}{12}\times\frac{2}{11} + \frac{5}{12}\times\frac{4}{11} + \frac{4}{12}\times\frac{3}{11}$

$= \frac{6}{132} + \frac{20}{132} + \frac{12}{132}$

$= \frac{38}{132}$

$= \frac{19}{66}$

## Probabilities on a probability tree diagram

- For 'AND' events, multiply the probabilities along the branches of each step.
- For 'OR' events, add the probabilities of separate branches together.

**EXERCISE 9.05** ANSWERS ON P. 619



### Selecting with and without replacement U F PS R C

EXAMPLE 10

**1** The positions of captain and vice-captain of a netball team are to be selected from Cate, Amal, Gemma, Josie, Evangeline and Rukshana. R C

**a** List the possible pairings of captain and vice-captain.

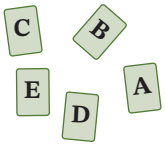
**b** What is the probability of Amal being captain or vice-captain?

**c** What is the probability of Evangeline becoming vice-captain?



Alamy Stock Photo/Tom Lindsey

**2** 2 cards are drawn from a set of cards labelled A, B, C, D and E. PS R C




**a** Make a list of all possible outcomes if the cards are drawn:

**i** with replacement **ii** without replacement.

**b** If the first card is replaced before the second card is drawn, find the probability that:

**i** both letters are the same

**ii** both letters are vowels

**iii** one letter is a vowel and the other is a consonant.

**c** If the first card is not replaced, find the probability that:

**i** both letters are vowels

**ii** one letter is a vowel and the other is a consonant

**iii** the first letter is a B or a D

**iv** the last letter is not A.

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**3** When staying at a hotel, David and Sarah can select one item from each course of a breakfast menu. R C

| 1st course | 2nd course |
|---|---|
| Cereal (C) | Bacon and eggs (B) |
| Fruit (F) | Ham and cheese croissants (H) |
| Yoghurt (Y) | Pancakes (P) |
| | Sausages and tomatoes (S) |
| | Toast and jam (T) |

**a** Copy and complete the table to list all the different 2-course breakfasts available.

| | | 2nd course | | | | |
|---|---|---|---|---|---|---|
| | | B | H | P | S | T |
| 1st course | C | | | | | |
| | F | | | | | |
| | Y | | | | | |

**b** If one of the combinations of breakfasts is chosen at random, what is the probability that it includes:

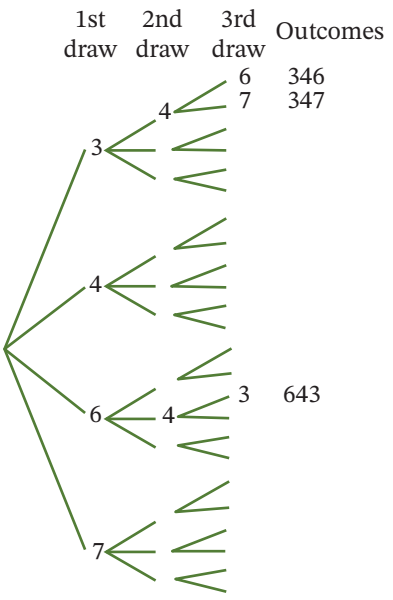
**i** fruit?

**ii** cereal but not bacon and eggs?

**iii** fruit and croissants?

**4** The numbers 3, 4, 6 and 7 are written on separate cards and placed in a bag. 3 cards are drawn at random without replacement to form a 3-digit number. PS R C

**a** Copy and complete the tree diagram to show all 24 possible outcomes.




**b** Find the probability of forming a number:

**i** that is even

**ii** greater than 400

**iii** between 400 and 700

**iv** that is even and greater than 400.

**5** The numbers 4, 5 and 9 are written on separate cards and placed in a bag. 3 cards are drawn at random with replacement to form a 3-digit number. PS R C

**a** Use a tree diagram to show all 27 possible outcomes.

**b** Find the probability of forming a number:

**i** with all digits the same

**ii** that is odd

**iii** less than 500

**iv** with all digits different.

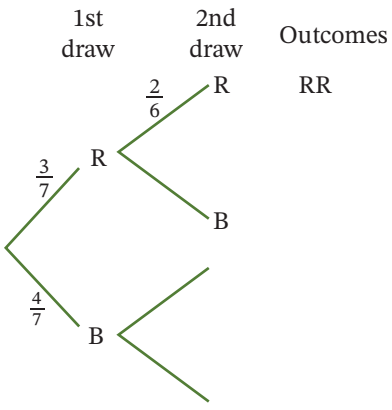
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EXAMPLE 11

9.05

**6** A bag contains 3 red and 4 blue marbles. 2 marbles are taken out of the bag without replacement. R C




**a** Copy and complete the probability tree diagram to list all possible outcomes.

**b** Find the probability of drawing:

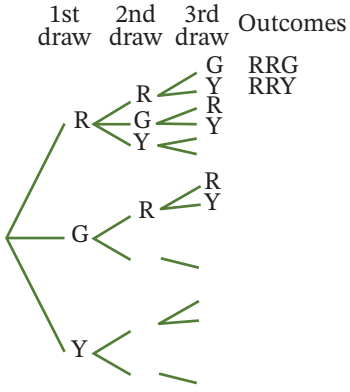
**i** 2 red marbles

**ii** 2 blue marbles

**iii** a blue and a red marble

**iv** at least one blue marble.

**7** A bag contains 2 red marbles, 1 green marble and 1 yellow marble. 3 marbles are drawn from the bag at random without replacement. PS R C




**a** Copy and complete the probability tree diagram to list the sample space.

**b** Find the probability of drawing:

**i** 2 red marbles

**ii** a red, green, and red marble in that order

**iii** at least one red marble.

**8** A family has 3 children. PS R C

**a** Use a tree diagram to list all possible outcomes in the sample space.

**b** What is the probability that the family consists of:

**i** 3 boys?

**ii** 3 girls?

**iii** 2 girls and a boy?

**iv** a girl and then 2 boys?

**9** 2 ice cream flavours are selected from vanilla, strawberry and chocolate. PS R C

**a** Draw a tree diagram to show the sample space if the flavours are selected:

**i** with replacement (repetition allowed)

**ii** without replacement.

**b** If the flavours are selected with replacement, find the probability of selecting:

**i** 2 identical flavours

**ii** 2 different flavours

**iii** no vanilla

**iv** at least one strawberry flavour.

**c** If the flavours are selected without replacement, find the probability of selecting:

**i** 2 identical flavours

**ii** 2 different flavours

**iii** no vanilla

**iv** at least one strawberry flavour.

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**10** For a long weekend (Saturday to Monday), the probability of rain on each day is $\frac{2}{5}$. R C

**a** Draw a probability tree diagram to show the possible outcomes for the 3 days. Label each branch of the tree diagram with its probability.

**b** For the long weekend, calculate the probability of:

**i** exactly 2 rainy days

**ii** no rainy days

**iii** exactly one rainy day

**iv** at least one rainy day.

**11** James, Max and Phoebe all play tennis. James serves an ace 4 out of 5 times on average, Max can serve an ace 3 out of 4 times and Phoebe 2 out of 3. They each complete a serve at the same time. R C

**a** Draw a probability tree diagram representing the possible outcomes for James, Max and Phoebe.

**b** What is the probability that:

**i** they all serve an ace?

**ii** none of them serve an ace?

**iii** James serves an ace but the other 2 players do not?

**iv** only one of the players serves an ace?

**12** Yen has 4 tickets in a raffle where 100 tickets have been sold. 3 tickets will be drawn without replacement for 1st, 2nd and 3rd prize. R C

**a** Draw a probability tree diagram for winning 1st, 2nd or 3rd prize.

**b** What is the probability of winning:

**i** 1st prize?

**ii** 2nd prize but no other prizes?

**iii** exactly one of the prizes?

**iv** none of the prizes?

## Investigation

### Dependent or independent?

Work in pairs.

*You will need*: a coin, 3 blue pens and 2 red pens.

**1 a i** Toss a coin and record the outcome (head or tail).

**ii** What is the probability of obtaining your outcome?

**b i** Toss the coin a second time and record the outcome.

**ii** What is the probability of obtaining the second outcome?

**c** Is the outcome of the second toss affected by the outcome of the first toss? Is the probability of the second outcome independent or dependent on the first outcome? Justify your answer.

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**2** **a** Copy this table.

| With replacement | 1st draw | 2nd draw |
|---|---|---|
| Blue | | |
| Red | | |
| | 40 | 40 |

**b** Put 3 blue pens and 2 red pens in a bag. Randomly draw a pen from the bag and record its colour.

**c** Put back the pen you drew and shake the bag. Draw a pen again and record its colour.

**d** Repeat the procedure from parts **a** and **b** 40 times and record the totals of each outcome in the table.

**e** Use your results to find:

**i** $P$(blue pen drawn first) **ii** $P$(blue pen drawn second).

**f** **i** Are your 2 results for part **e** the same?

**ii** Would you expect the results to be the same? Give reasons.

**g** Is the outcome of the second draw dependent on the outcome of the first draw?

**3** **a** Copy this table.

| Without replacement | 1st draw | 2nd draw |
|---|---|---|
| Blue | | |
| Red | | |
| | 40 | 40 |

**b** Again, place 3 blue pens and 2 red pens in a bag. Randomly draw a pen from the bag and record its colour.

**c** Do not return the pen, shake the bag and draw a second pen, recording its colour.

**d** Repeat the procedure from parts **a** and **b** 40 times and record the totals of each outcome in the table.

**e** Use your results to find:

**i** $P$(blue pen drawn first) **ii** $P$(blue pen drawn second).

**f** **i** Are your 2 results for part **e** the same?

**ii** Would you expect your results to be the same? Give reasons.

**g** Is the outcome of the second draw dependent on the outcome of the first draw? Compare your results with those of other students in your class.



# Dependent and independent events 9.06

Two events are **independent** if the outcome of one event **does not affect** the outcome of the other event. So, one event occurring does not change the probability of the other event. For example, if a coin and a die are tossed together, the 2 events are independent as the outcome on the coin does not affect the outcome on the die. Wearing blue socks and passing a driving test are also independent events.

Two events are **dependent** if the outcome of one event **does affect** the outcome of the other event. So, one event occurring changes the probability of the other event occurring. For example, when selecting 2 coloured pencils from a pencil case without replacement, the 2 events are dependent because the outcome of the second draw is affected by the outcome of the first draw. Raining on a school sports day and sport being cancelled are also dependent events.

**Videos**
The Monty Hall problem
The birthday paradox
The card counter
The prisoner's dilemma
Bayesian robots

## Example 12

A coin and a die are tossed together.

**a** List the outcomes in the sample space.

**b** Find:

**i** $P(\text{head on the coin})$ **ii** $P(\text{even number on the die})$

**iii** $P(\text{head and even number})$.

**c** Is $P(\text{head and even number}) = P(\text{head}) \times P(\text{even number})$?

**d** Are the 2 events dependent or independent?

### Solution

**a** The outcomes are H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5 and T6.

**b** **i** $P(\text{head}) = \frac{1}{2}$ **ii** $P(\text{even}) = \frac{3}{6} = \frac{1}{2}$

**iii** $P(\text{head and even}) = \frac{3}{12} = \frac{1}{4}$ H2, H4 and H6

**c** Yes, since $P(\text{head and even}) = \frac{1}{4}$ and $P(\text{head}) \times P(\text{even}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

**d** The 2 events are independent since the outcome when tossing a coin does not affect the outcome when rolling a die.

## The product rule for independent events

Two events are **independent** if the outcome of one event does not affect the outcome of the other event.

If $A$ and $B$ are 2 independent events, then $P(A \text{ and } B) = P(A) \times P(B)$.

## Example 13

A bag contains 3 brown chocolates and 1 white chocolate. 2 chocolates are drawn from the bag, without replacement.

**a** Find the probability of:

**i** selecting a brown chocolate with the first draw

**ii** selecting a brown chocolate with the second draw if the first chocolate was brown.

**b** Are selecting a brown chocolate with the first and second draws dependent or independent?

### Solution

**a** **i** $P(\text{brown on the first draw}) = \frac{3}{4}$

**ii** After drawing a brown chocolate, there are 3 chocolates left, of which 2 are brown.

$\therefore P(\text{brown on the second draw}) = \frac{2}{3}$

**b** The bag contains 2 brown chocolates and 1 white chocolate for the second draw, so $P(\text{brown})$ decreases from $\frac{3}{4}$ to $\frac{2}{3}$. The second event is dependent on the first event.

**EXERCISE 9.06** ANSWERS ON P. 621

## Dependent and independent events U F PS R C

**1** State whether each pair of events are dependent or independent. R C

- **a** Rolling 4 on a die and rolling an even number on another die
- **b** Rolling 6 on a die and rolling 6 again on the same die
- **c** Training hard at soccer and winning a soccer match
- **d** Drawing a red ball from a bag containing red and blue balls, replacing it and then drawing a blue ball from the bag
- **e** Electing a team captain from a group of players and then electing a vice-captain from the same group
- **f** Tossing 2 coins and obtaining a head on the first coin and a head on the second coin
- **g** Finding $20 on the street and getting a phone call from your parent

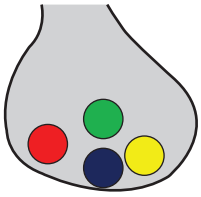
**2** In Lotto, 6 balls are drawn without replacement. Are the events of drawing each of the balls dependent or independent events? Give reasons. R C

**3** A coin is tossed 3 times and the result is heads each time. R C

- **a** Are each of the 3 coin tosses dependent or independent events?
- **b** The coin is tossed a 4th time. What is the probability of obtaining a head on the 4th toss?

EXAMPLE 12

**4** A normal die is rolled and a marble is drawn from a bag containing a yellow marble, green marble, blue marble and red marble. R C





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- **a** Find the probability of:
  - **i** rolling a number less than 3 with the die
  - **ii** drawing a green marble from the bag.
- **b** List the outcomes for rolling the die and drawing a marble from the bag.
- **c** What is the probability of rolling a number less than 3 and drawing a green marble?
- **d** Is $P(\text{rolling a number less than 3}) \times P(\text{drawing a green marble}) = P(\text{a number less than 3 and a green marble})$?
- **e** Are the events of rolling a number less than 3 and drawing a green marble dependent or independent?

**5** A red die and a blue die are rolled. R

- **a** Find:
  - **i** $P(5 \text{ on the red die})$
  - **ii** $P(\text{an even number on the blue die})$.
- **b** Hence find $P(5 \text{ on the red die AND an even number on the blue die})$.

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EXAMPLE 13

**6** A bag contains 5 red balls and 4 yellow balls. 2 balls are drawn at random without replacement. R C

**a** What is the probability of drawing a red ball first?

**b** What is the probability of drawing a red ball on the second draw if the first ball was red?

**c** Are the 2 events dependent or independent? Give reasons.

**7** A bag contains 5 yellow counters and 3 red counters. 2 draws are made with no replacement. Find the probability of drawing: R C

**a** **i** a yellow counter on the first draw

**ii** a yellow counter on the second draw after a yellow counter was drawn with the first draw

**b** **i** a red counter on the first draw

**ii** a yellow counter on the second draw after a red counter was drawn on the first draw

**c** **i** a yellow counter on the first draw

**ii** a red counter on the second draw after a yellow counter was drawn on the first draw

**d** **i** a red counter on the first draw

**ii** a red counter on the second draw after a red counter was drawn on the first draw.

**8** 3 children in a family are all girls. What is the probability that the next child in this family will be a girl? R

**9** A pair of dice are rolled. Use the product rule to find the probability of rolling: PS R

**a** a 6 on the first die and an odd number on the second die

**b** a 1 on both dice

**c** a prime number on the first die and a factor of 6 on the second die

**d** an even number on the first die and a number between 2 and 5 on the second die.

**10** The probability that Natalie wins her tennis match is $\frac{13}{20}$. The chance that she will pass her Science exam is $\frac{4}{5}$. What is the probability that Natalie: PS R

**a** wins her tennis match and passes her Science exam?

**b** loses her tennis match and passes her Science exam?

**11** The probability that it will rain on Saturday is 0.6. The probability that it will rain on Sunday is 0.25. What is the probability that it will rain: PS R

**a** on both Saturday and Sunday?

**b** on Sunday only?

**c** on neither Saturday nor Sunday?

**12** A main road has 3 sets of traffic lights, each with a probability of 0.7 of being green. Use the product rule to find the probability of: PS R

**a** green on the first 2 lights, red (or amber) on the 3rd light

**b** green on all 3 lights

**c** red (or amber) on all 3 lights

**d** only one of the 3 lights being green.

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## Did you know?

### Lotteries and Lotto

A lottery is a game of chance in which numbered tickets are drawn from tickets that have been sold. Lotteries were introduced by the State Government to raise money for hospitals. The first lottery was drawn on 20 August, 1931 with a first prize of £5000.

Lotteries have been used to celebrate special events and to help finance special projects. The Opera House Lottery, which commenced selling on 25 November 1957, was used to finance the construction of the Sydney Opera House.



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Other games of chance have been introduced, including Lotto (1979), Instant Scratchies (1982), OzLotto (1994) and Powerball (1996).

**Research the probability of winning Lotto, OZ lotto and Powerball.**

## Power plus ANSWERS ON P. 621

**1** Students at Nelson Secondary College were surveyed about which sport they like to watch and what type of movies they like to see.

| | Horror/Drama | Fantasy | Comedy | Action |
|---|---|---|---|---|
| Football | 23 | 34 | 30 | 48 |
| Cricket | 25 | 12 | 45 | 34 |
| Tennis | 8 | 12 | 32 | 17 |

**a** How many students were surveyed?

**b** If a student is selected at random, what is the probability that the student likes to watch:

**i** horror/drama movies?

**ii** comedy and football?

**iii** tennis, but not fantasy?

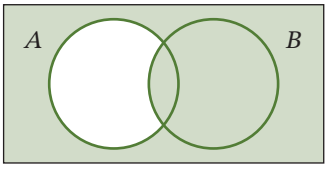
**iv** action, but not cricket or tennis?

**c** Given that a student likes to watch football, find the probability that the student also likes to watch action movies.

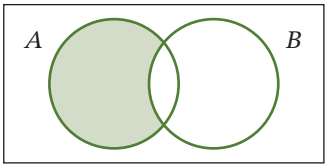
**d** Of the students who like comedy, what is the probability that they also like to watch cricket?

**2** Use set notation to describe the shaded region in each Venn diagram.

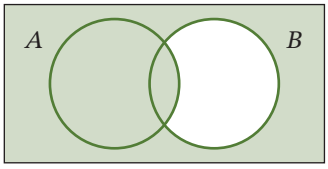
**a**



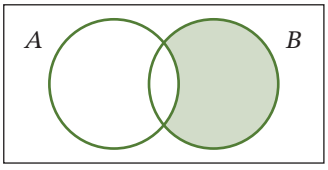

**b**



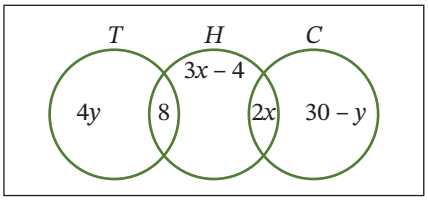

**c**




**d**




**3** 79 students play at least one of the following sports: tennis ($T$), hockey ($H$) and cricket ($C$). The Venn diagram shows the number of students involved in each sport in algebraic form.




**a** Write a formula for $y$ in terms of $x$.

**b** Given that the number of students who played tennis is 8 times as many as those who played both hockey and cricket, find $x$ and $y$.

**c** Find the number of students who only played hockey.

**4** 2 hair ties are drawn without replacement from a bag containing 3 purple and 2 red hair ties.

**a** Copy and complete the tree diagram to show the sample space.

**b** Given that the first hair tie drawn is purple, what is the probability that the second hair tie will be:

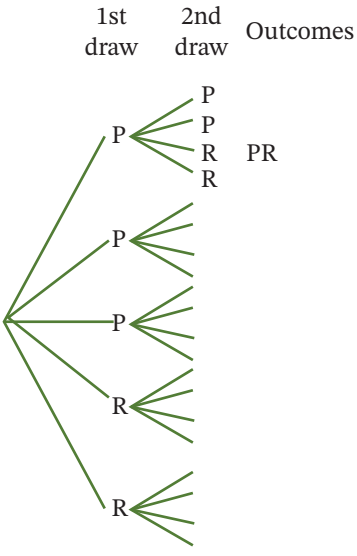
**i** red? **ii** purple?

**c** Given that the first hair tie drawn is red, find the probability that the second hair tie is:

**i** red **ii** purple.

**d** Given that the drawn hair ties are the same colour, what is the probability that they are:

**i** both red? **ii** both purple?





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# CHAPTER 9 REVIEW

## Language of maths

| | | |
|---|---|---|
| complementary | compound event | dependent event |
| die/dice | event | expected frequency |
| expected probability | independent event | intersection |
| mutually exclusive | observed probability | probability tree diagram |
| random | relative frequency | sample space |
| set | simulation | subset |
| tree diagram | two-stage experiment | two-way table |
| union | Venn diagram | with/without replacement |

**1** What is the meaning of **expected frequency**?

**2** Which word means to imitate or recreate a chance experiment, usually by a computer model?

**3** On a **Venn diagram**, what does the rectangle represent?

**4** Give an example of **dependent events**.

**5** When are **tree diagrams** used in probability?

**6** For 2 events A and B, what is the difference between '**A or B**' and '**A and B**'?

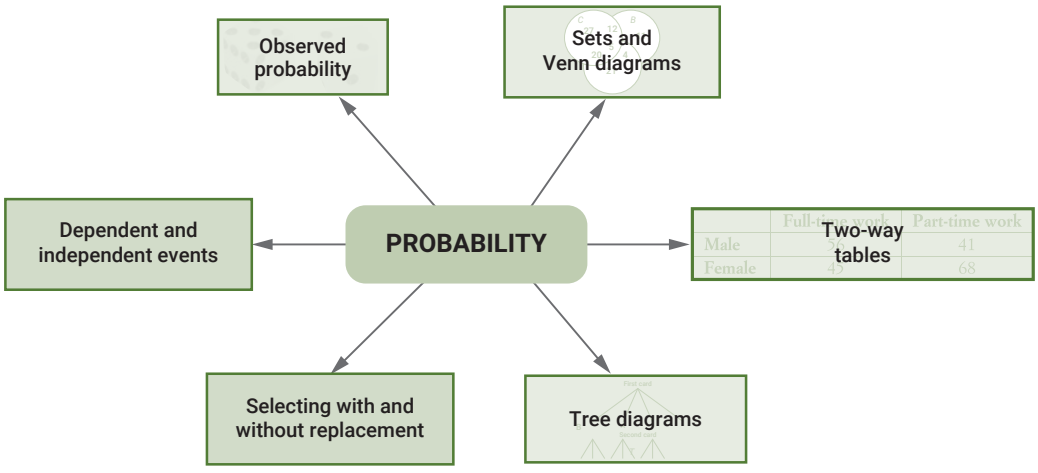
**Quiz**
Language of maths 9

**Puzzle**
Probability crossword

## Topic summary

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.

**Worksheet**
Mind map: Probability



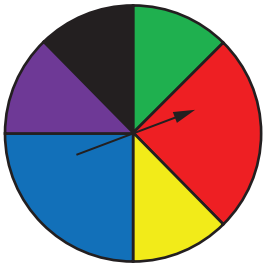
Quiz
Test yourself 9

# TEST YOURSELF 9

ANSWERS ON P. 622

9.01

**1** For the spinner shown, the red and blue sectors are twice as large as the other sectors. Rafiya spun the spinner 100 times and her results are shown in the table.



| Outcome | Frequency |
|---|---|
| Red | 22 |
| Blue | 29 |
| Yellow | 10 |
| Purple | 12 |
| Black | 13 |
| Green | 14 |

**a** What is the observed probability (as a decimal) of the arrow stopping on:

**i** red? **ii** blue? **iii** yellow? **iv** green?

**b** What is the expected probability (as a decimal) of the arrow stopping on:

**i** red? **ii** blue? **iii** yellow? **iv** green?

**c** Are the observed and expected probabilities similar?

**d** What is the observed probability of the arrow stopping at purple or black?

**e** What is the expected frequency of a colour that is not purple or black? How does this compare with Rafiya's observed frequency?

9.01

**2** 3 coins are tossed 150 times and the number of heads at each trial is recorded in the table.

| Number of heads | Frequency |
|---|---|
| 0 | 20 |
| 1 | 53 |
| 2 | 64 |
| 3 | 13 |

**a** Find correct to 3 decimal places the relative frequency of tossing:

**i** one head **ii** 2 heads

**iii** 3 heads **iv** at least 2 heads.

**b** Find correct to 3 decimal places the observed probability of:

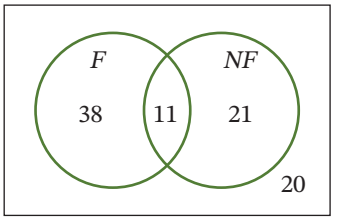
**i** at least one head **ii** 3 tails.

**c** Are the answers in part **b** the same or different? Explain why.

PATH

9.02

**3** People at a train station were asked whether they prefer to read fiction ($F$) or non-fiction ($NF$) books. The results are shown in the Venn diagram.




**a** How many people were surveyed?

**b** How many people are in the set:

**i** $\overline{NF}$? **ii** $F \cup NF$? **iii** $\overline{F} \cap NF$?

**c** Find the probability of selecting a person from this group who only reads fiction books.

**d** What is the probability of selecting a person who doesn't read fiction or non-fiction books?

**e** Of the people who read fiction books, find the probability that they read non-fiction books.

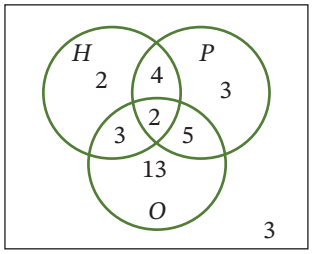
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PATH

**4** This Venn diagram shows the results of a survey on the types of music that school students listen to – Hip Hop/Rap ($H$), Pop ($P$) and Other ($O$), which includes R&B, Dance, Metal, Rock and Independent.




9.02

**a** How many students were surveyed?

**b** How many students are in the set:

**i** $H \cap O$? **ii** $H \cup P$? **iii** $(O \cap P) \cup H$?

**c** Find the probability of selecting a student who likes to listen to all types of music.

**d** What is the probability of selecting a student who listens to

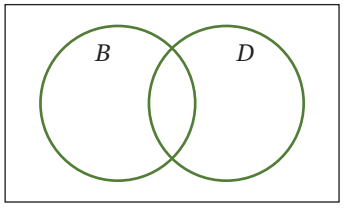
**i** Hip Hop/Rap and Pop?

**ii** Hip Hop/Rap or Pop?

**iii** Pop music only?

**e** Why are 3 students in the rectangle, but not in the circles?

**5** Of 20 people in an office, 6 have blue eyes ($B$), 8 have dark hair ($D$) and 3 have blue eyes and dark hair.




9.02

**a** Copy and complete the Venn diagram to show the given information.

**b** What is the probability of selecting a person at random from the office who has:

**i** blue eyes only? **ii** dark hair?

**iii** blue eyes and dark hair? **iv** hair that is not dark?

**c** What is the probability of selecting a person at random who has neither blue eyes nor dark hair?

**6** Students were asked what type of activities they would like to do on a camp.

9.03

| | Hiking | Rock climbing | Kayaking |
|---|---|---|---|
| Boys | 25 | 38 | 47 |
| Girls | 45 | 23 | 22 |

**a** How many students were surveyed?

**b** Find the probability (as a decimal) that a student selected at random:

**i** likes rock climbing

**ii** likes kayaking and is a girl

**iii** is a girl who likes hiking

**iv** is a boy who likes rock climbing or kayaking.

**c** If a boy is selected at random, what is the probability that his favourite activity is hiking?

**d** If a student who likes kayaking is chosen, what is the probability that the student is:

**i** a boy? **ii** a girl?

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9.04

**7** A pair of dice are rolled and the sum of the 2 numbers is calculated.
Find the probability of rolling a sum of 6. Select the correct answer **A**, **B**, **C** or **D**.

**A** $\frac{1}{5}$ **B** $\frac{1}{9}$ **C** $\frac{5}{6}$ **D** $\frac{5}{36}$

**8** A pair of 4-sided dice (numbered 1, 2, 3 and 4) are rolled.

| | | 1st die | | | |
|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 |
| 2nd die | 1 | | | | |
| | 2 | | 2, 2 | | |
| | 3 | | | | |
| | 4 | | | 3, 4 | |

**a** Copy and complete this table.

**b** How many possible outcomes are there?

**c** Find the probability of rolling:

**i** one odd and one even number

**ii** 2 even numbers

**iii** at least one 3

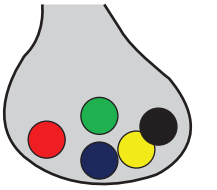
**iv** 2 numbers less than 3

**v** a double

**vi** 2 numbers so that the first number is odd.

9.05

**9** 2 marbles are drawn from a bag containing a red, a blue, a green, a yellow and a black marble.



**a** Make up a list to show all the possible outcomes if the marbles are taken:

**i** with replacement

**ii** without replacement.

**b** If the first marble is replaced before the second marble is drawn, find the probability of drawing:

**i** 2 red marbles

**ii** 2 marbles of the same colour

**iii** a yellow and a black marble

**iv** at least one green marble.

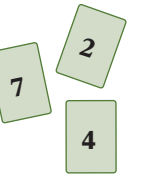
**c** If the first marble is not replaced, find the probability of drawing:

**i** a green marble and a yellow marble

**ii** no red marbles.

9.05

**10** The numbers 2, 4, and 7 are written on separate cards and placed in a bag.
3 cards are drawn at random to form a 3-digit number.




**a** Make up a tree diagram to list the sample space if the cards are drawn:

**i** with replacement

**ii** without replacement.

**b** If the cards are drawn with replacement, find the probability of forming:

**i** an even number

**ii** a number less than 400

**iii** the numbers 222, 444, or 777

**iv** an odd number greater than 400.

**c** If the cards are drawn without replacement, find the probability of forming:

**i** an odd number

**ii** a number greater than 400

**iii** a number beginning with 7

**iv** a number divisible by 4.

**11** A fruit bowl contains 2 apples and 3 oranges. Two pieces of fruit are selected at random, without replacement. 9.05

**a** Draw a probability tree diagram showing all probabilities on the branches.

**b** What is the probability of selecting:

**i** 2 apples? **ii** 2 different fruits? **iii** at least 1 orange?

**12** State whether each pair of events are dependent or independent. 9.06

**a** Tossing a tail from a coin and then a head from the same coin.

**b** Drawing a ticket in a raffle and winning a first prize and then drawing a second ticket and winning a second prize.

**c** Electing a president for a cricket club and then electing the vice-president of the cricket club.

**d** A family's first 3 children are girls and then the 4th child is also a girl.

**e** Rain in your town or city today and a car accident.

**f** The Roosters winning a football match and you winning Lotto.